

FACTOR-SUM RETURNS AND FIXED-TAIL PELL FIBRES FOR RATIONAL ERDŐS-STRAUS TRACES

THE CLANKERS

ABSTRACT. An original middle trace at a prime-square distinguished denominator can have integral tail sum while every full gate at its shell, and every fixed-word or fixed-ray residual-divisor return, fails. We prove an explicit different return preserving the sum of two factors of the distinguished denominator. Its entire domain and inverse are given by one quadratic equation. The prime-square specialization is Pell, with an explicit infinite integer family and independently certified hard-prime examples. A square-zero coordinate also allows genuinely mixed E/M cancellation inside the unchanged original exponent box. A second exact Pell lattice classifies every integral target that retains the numerical tail sum. For every proper prime-square trace, no such target can have residual a proper divisor of the source residual. At $p = 67369$ the complete fibre is empty for every proper original trace sum. No universal source-existence assertion is made.

Let $p \equiv 1 \pmod{4}$ be prime, $p/4 < a < p/2$, $R = 4a - p$, and $u \mid a^2$. With $g = \gcd(a, u)$ and $(h, r, s) = (g^2/u, u/g, a/g)$, retain

$$\begin{aligned} E : & (a, h\kappa, phr\kappa), & \kappa &= (pr + s)/R, \\ M : & (a, phs\lambda, phr\lambda), & \lambda &= (r + s)/R. \end{aligned}$$

These are rational reciprocal identities. Their full integer gates are $R \mid 4u + 1$ and $R \mid p + 4u$; integral tail sum requires their squares instead. The ordered divisor inverses are $a^2/(Ry - pa)$ and $pa^2/(Ry - pa)$. These inherited Type-I/II coordinates are reviewed in Elsholtz and Tao [1, §2]. We retain the complete box $0 \leq v_\ell(u) \leq 2v_\ell(a)$.

Lemma 1 (Prime-square terminal trace). *Suppose $p \equiv 1 \pmod{24}$ and $a = q^2$, $q > 3$ prime. Every M trace has $u = q$ or q^3 , $R \mid (q + 1)^2$. If $R \nmid q + 1$, its whole original shell has no full E/M state. Neither word, nor either primitive M ray, admits any proper residual-divisor return preserving that word or ray.*

Proof. One has $R \equiv 3 \pmod{24}$. Reduction of the square gates modulo three forces $q \equiv 2 \pmod{3}$ and odd word exponent; hence only q, q^3 occur. The two M square gates are equivalent to $R \mid (q + 1)^2$. Write $K = \prod_{\ell \in \mathbb{N}} \ell^{\lceil e/2 \rceil}$. If an E trace also occurred, $q \equiv -1 \pmod{K}$ would give $K \mid 4q^f + 1 \equiv -3 \pmod{K}$ for its odd f . This forces $R = 3$, incompatible with properness. Both M words are nonintegral by assumption.

A deletion to R/k retaining $u = q$ or either primitive M ray requires $k \mid R$, $k > 1$, $k \equiv 1 \pmod{4q}$. Such k divides $(q + 1)^2$. Its complement $v = (q + 1)^2/k$ exceeds one (otherwise k is even), and is one modulo q . Writing $k = 1 + qA$, $v = 1 + qB$ gives $A \geq 4$, $B \geq 1$ and $qAB + A + B = q + 2$, impossible. For the other word q^3 , the word-preservation modulus is $4q^2 > R$, which also excludes every proper k . $\square$

Theorem 2 (Factor-sum arrow). *Take an original M trace with $a = hs$, $u = h$, $r = 1$, and $R = 4hs - p = \delta t^2$, where δ is squarefree. Thus $\delta \equiv 3 \pmod{4}$ and $\delta t \mid s + 1$. Its factor-sum-preserving returns with residual δ are exactly*

$$h' = h - \delta w, \quad s' = s + \delta w, \quad \delta w^2 + (s - h)w = (t^2 - 1)/4, \quad w \in \mathbb{Z}.$$

Every integer root gives positive integers h', s' , an available word $u' = h' \mid (h's')^2$, and the ordered identity

$$\boxed{\frac{4}{p} = \frac{1}{h's'} + \frac{1}{ph's'\lambda'} + \frac{1}{ph'\lambda'}, \quad \lambda' = (s' + 1)/\delta.}$$

For a proper trace, $h's' < hs$.

Proof. Expansion gives $h's' = hs + \delta w(h - s) - \delta^2 w^2$. Thus the quadratic is exactly $4h's' - p = \delta$. The target product $(p + \delta)/4$ and factor sum $h + s$ are positive, proving positivity of both factors. The original $\delta \mid s + 1$ gives $\delta \mid s' + 1$, so the target is the full M marking $(h', 1, s', \lambda')$. Multiplying its reciprocal sum by $ph's'\lambda'$ gives $p\lambda' + s' + 1 = (p + \delta)\lambda' = 4h's'\lambda'$. The first-half range follows from $3 \leq \delta < p$. Properness implies $t > 1$, hence $hs - h's' = \delta(t^2 - 1)/4 > 0$. $\square$

The map is an explicit finite algorithm. Its discriminant is $(h - s)^2 + R - \delta$; test its integer square root and divisibility by 2δ . There are at most two possible w . Its full incoming fibre at a marked target (p, δ, h', s') consists of integers

$$-h'/\delta < w < s'/\delta,$$

for which $1 + 4w(s' - h') - 4\delta w^2 = t^2$ is a positive odd square, $\delta t^2 < p$, and $\delta t \mid s' - \delta w + 1$. Recover $h = h' + \delta w$, $s = s' - \delta w$, $R = \delta t^2$, and $u = h$. These tests are necessary and sufficient by the same expansion; retaining w gives both inverse compositions. The other M orientation is restored using the new word's complement in the new square-divisor box.

Pell specialization and an integer family. For $h = s = q$, the quadratic is precisely

$$t^2 - 4\delta w^2 = 1.$$

For any positive solution, any squarefree $\delta \equiv 3 \pmod{4}$ and $k \geq 1$, put

$$q = \delta tk - 1, \quad p = 4q^2 - \delta t^2, \quad h' = q - \delta w, \quad s' = q + \delta w, \quad \lambda' = tk + w.$$

All quantities in the boxed reciprocal identity are positive integers. Indeed $q \geq \delta t - 1$ gives $p > \delta t^2$; Pell gives $t > 2\sqrt{\delta}w$, hence $q > \delta w$; and $\delta t \mid q + 1$ supplies the original trace. Its exact reduced tail denominator is $t/\gcd(t, k)$. This proves an integer identity without asserting that either quadratic parameter is prime.

For $\delta = 3$, the recurrence

$$(t_{n+1}, w_{n+1}) = (7t_n + 24w_n, 2t_n + 7w_n), \quad (t_0, w_0) = (1, 0)$$

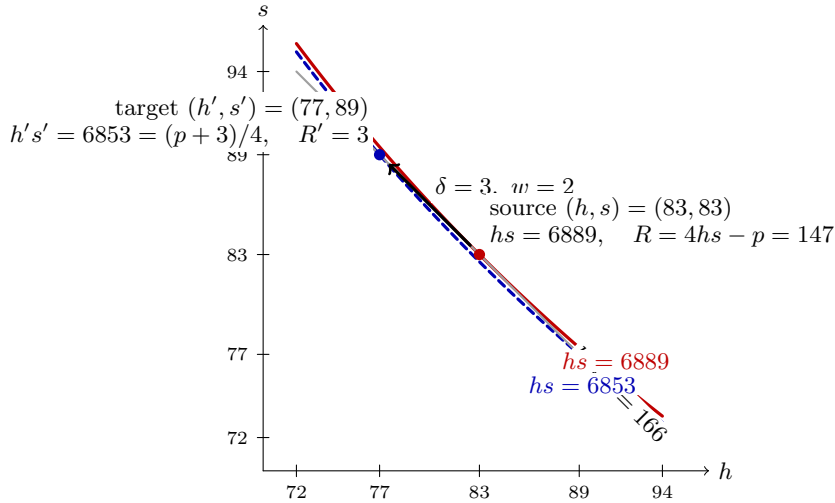
preserves $t^2 - 12w^2 = 1$ and yields infinitely many such integer inputs. The first two certified prime instances are

t	w	k	q	p	(h', s', λ')
7	2	4	83	27409	(77, 89, 30)
97	28	4	1163	5382049	(1079, 1247, 416).

At the first, $R = 147$, $a = 6889$ is a terminal prime-square trace as above, but the new word is 77, the new primitive ray is $(1, 89)$, and

$$\frac{4}{27409} = \frac{1}{6853} + \frac{1}{5635016310} + \frac{1}{63314790}.$$

The construction changes both invariants held fixed by the failed maps. No infinitude of prime values in the Pell family is claimed.



$p = 27409$, $r = 1$, and $(h, s) \mapsto (h - \delta w, s + \delta w)$ sends $(83, 83)$ to $(77, 89)$ while retaining the factor sum $h + s = 166$. Both product levels and both residuals are displayed; the arrow does not identify the two hyperbolas.

FIGURE 1. The exact factor-sum arrow $(h, s) \mapsto (h - \delta w, s + \delta w)$ on the first Pell-family example. The source lies on $hs = 6889$ and the target lies on $hs = 6853$; their common line is $h + s = 166$.

An exact mixed-channel arithmetic defect. Write $R = KD$, where $K = \prod \ell^{\lceil v_\ell(R)/2 \rceil}$ and $D = R/K$. The ideal $K\mathbb{Z}/R\mathbb{Z}$ has square zero. With $\beta_\ell = v_\ell(u) - v_\ell(a)$ and $\eta = 1$ for E, 0 for M, a trace has the unique coordinate

$$p^\eta u/a = -1 + Kc \pmod{R}, \quad c \in \mathbb{Z}/D.$$

Its full gate is $c = 0$, and its tail denominator is $D/\gcd(D, c)$. For traces indexed by j , any integers n_j with odd sum N , exact channel exponent $\sum n_j \eta_j \in \{0, 1\}$, and retained bounded vector $\sum n_j \beta^{(j)}$ produce the original word

$$u_* = a^{1-N} \prod u_j^{n_j}, \quad c_* = \sum n_j c_j \pmod{D}.$$

The proof is the finite identity $\prod (1 - Kc_j)^{n_j} = 1 - K \sum n_j c_j$, valid even for negative powers. The exponent-box condition is indispensable: a unit residue alone is not a new integer divisor.

At $(p, a, R) = (48049, 12019, 27)$, the proper trace words

$$(E, 29189, c = 1), \quad (E, 119, c = 2), \quad (M, 707, c = 1)$$

have no individual fixed-word or fixed-ray divisor deletion. The signed combination $E_1 - E_2 + M$ is nevertheless an available M word 173417, with $c = 0$, and gives

$$\frac{4}{48049} = \frac{1}{12019} + \frac{1}{22871324} + \frac{1}{330000532}.$$

The full paper partitions every original box into finite cyclic coefficient polynomials and proves a sharp saturation threshold, without replacing bounded exponents by their generated subgroup.

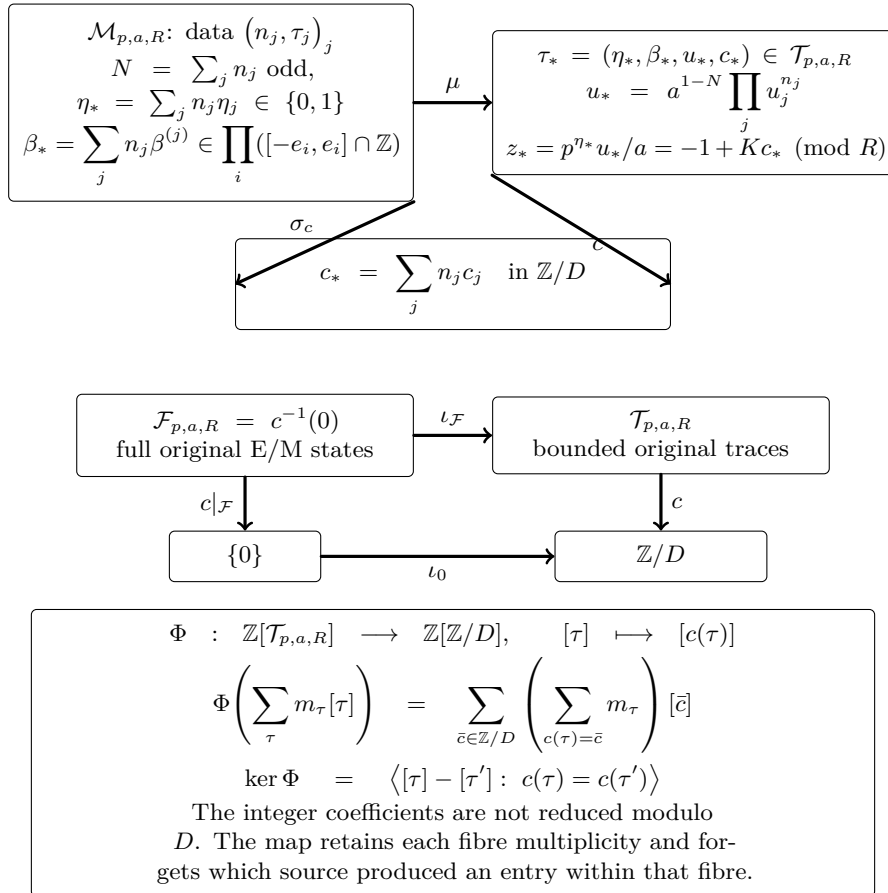


FIGURE 2. Typed maps for the defect equation $p^\eta u/a = -1 + Kc \pmod{R}$. In the upper triangle, $\sigma_c((n_j, \tau_j)_j) = \sum_j n_j c_j$, and the triangle commutes by the mixed integer-exponent law. The middle square identifies the full states as the zero fibre. The lower map states the exact kernel and the information lost when original trace words are replaced by their defect-bin coefficient vector.

The distinct fixed-tail-sum fibre. The factor-sum arrow above retains $h + s$. Fixing instead the numerical tail sum $Y + Z = N$ gives a different exact problem. Put

$$\mathcal{D} = N(N - p), \quad \rho = 4A - p, \quad \Delta = Z - Y.$$

The ordered positive integer triples with $p/4 < A < p/2$ and

$$\frac{4}{p} = \frac{1}{A} + \frac{1}{Y} + \frac{1}{Z}, \quad Y + Z = N$$

are in bijection with

$$\rho \mid N, \quad 0 < \rho < p, \quad \rho \equiv -p \pmod{4}, \quad \Delta^2 = \mathcal{D} - \frac{Np^2}{\rho}, \quad |\Delta| < N, \quad \Delta \equiv N \pmod{2}.$$

The inverse is

$$(A, Y, Z) = \left(\frac{p + \rho}{4}, \frac{N - \Delta}{2}, \frac{N + \Delta}{2} \right).$$

Indeed the reciprocal identity gives $YZ = Np(p + \rho)/(4\rho)$, hence $\rho\Delta^2 = N((N - p)\rho - p^2)$. Since p is prime and $0 < \rho < p$, this forces $\rho \mid N$. The displayed inverse proves sufficiency.

Writing $\rho = \delta v^2$ and $X = v\Delta$ gives the exact lattice equation

$$X^2 - \mathcal{D}v^2 = -\frac{Np^2}{\delta}, \quad \delta v^2 \mid N, \quad X \equiv Nv \pmod{2v}.$$

The divisibility and congruence gates are essential. Equivalently, if $B = \rho Y - pA$, $C = \rho Z - pA$, $w = B - C$, then

$$(2\mathcal{D}\rho - p^2N)^2 - \mathcal{D}(2w)^2 = p^4N^2,$$

but integer tails additionally require $\rho \mid w$.

Theorem 3 (Prime-square fixed-sum divisor obstruction). *Let p, q be odd primes with $p \equiv 1 \pmod{4}$ and $2q^2 < p < 4q^2$, and set $R = 4q^2 - p$. Choose $u = q$ or q^3 and an E/M channel. If its prime-square source is a proper trace, then no positive integral target with the same numerical tail sum N has residual $S = 4A - p$ satisfying $S \mid R$ and $0 < S < R$.*

Proof. Put $G_E(u) = 4u + 1$ and $G_M(u) = p + 4u$. In the four cases define

$$\begin{array}{c|c|c} (E, q) & H = p + q & N = qH^2/R \\ (E, q^3) & H = pq + 1 & N = qH^2/R \\ (M, q), (M, q^3) & H = q + 1 & N = pqH^2/R. \end{array}$$

The identities

$$H = qG_E(q) - R, \quad H = G_E(q^3) - qR, \quad G_M(q) = 4qH - R, \quad G_M(q^3) = qR + pH$$

show that properness is $R \mid H^2$ and $R \nmid H$. Let

$$g = (R, H), \quad \eta = R/g, \quad F = H/g, \quad k_0 = g/\eta.$$

Then

$$R = k_0\eta^2, \quad H = k_0\eta F, \quad (\eta, F) = 1, \quad \eta \geq 3,$$

with $(pq, k_0F) = 1$, and

$$N = qk_0F^2 \quad (E), \quad N = pqk_0F^2 \quad (M), \quad (R, N) = k_0.$$

Suppose a target exists. The exact inequalities are

$$A = \frac{p + S}{4} < \frac{p + R}{4} = q^2 < \frac{p}{2}.$$

Moreover $S \equiv 3 \pmod{4}$. From $SYZ = NpA$ and $(S, pA) = 1$ one gets $S \mid N$, hence $S \mid k_0$. Factoring

$$(SY - pA)(SZ - pA) = (pA)^2$$

with the p -valuation pair retained shows, up to the E tail orientation, that every target is exactly one of

$$\begin{array}{c|c|c} E & S\kappa = pr + s & N = hS\kappa^2 \\ M & S\lambda = r + s & N = phS\lambda^2, \end{array} \quad A = hrs, \quad (r, s) = 1.$$

Thus the parity of $v_p(N)$ forces the target into the source channel. Writing $\theta = \kappa$ in E and $\theta = \lambda$ in M gives

$$q \frac{k_0}{S} F^2 = h\theta^2.$$

Since $q \nmid (k_0/S)F$ and $h \mid A < q^2$, one has $h = qh_0$. Hence

$$A = q(q - j), \quad 0 < j < q, \quad R - S = 4qj, \quad \frac{k_0}{S}F^2 = h_0\theta^2.$$

For (E, q) , the gate $S\kappa = pr + s > p$ contradicts

$$S\kappa \leq F\sqrt{Sk_0} \leq Fk_0 = \frac{p+q}{\eta} < p.$$

For either M orientation, $q = k_0\eta F - 1$; here k_0, η are odd and F is even. Moreover

$$k_0\eta(\eta - 4jF) = S - 4j.$$

Here $S - 4j$ is nonzero, while $0 < S < k_0\eta$ and $0 < 4j < (k_0\eta^2)/q < k_0\eta$ because $q \geq 2\eta - 1 > \eta$. This is impossible for a nonzero multiple of $k_0\eta$.

It remains to treat (E, q^3) . Write $F = nw$, $\kappa = mw$, $(m, n) = 1$. There is a positive τ such that

$$k_0/S = \tau m^2, \quad h_0 = \tau n^2.$$

With $\Gamma = \tau mn\eta$,

$$pq + 1 = k_0\eta F = \Gamma S\kappa = \Gamma(pr + s),$$

so $\Gamma s \equiv 1 \pmod{p}$. Also

$$0 < \tau ns = \frac{q-j}{nr} < q.$$

Finally $R = S\tau(m\eta)^2 < 2q^2$, and $S \geq 3$, so $m\eta < q$. Therefore $0 < \Gamma s < q^2 < p$, forcing $\Gamma s = 1$, contrary to $\Gamma s \geq \eta \geq 3$. $\square$

The divisor condition is essential to the theorem: it leaves open residuals $S < R$ with $S \nmid R$, and changing N is outside this fibre.

For the proper trace

$$(p, a, R, u, h, r, s) = (67369, 16849, 27, 1421, 29, 7, 83), \quad N = 586110300,$$

one has

$$(B, C, w) = (13459046189, 95731349, 13363314840), \quad w \equiv 18 \pmod{27}.$$

The only residual candidates are 3, 15, 75, 87, 435, 2175. The first has negative discriminant; the other discriminants are respectively 2 modulo 3, 5 modulo 7, 2 modulo 3, 3 modulo 7, and 2 modulo 3. Thus none is a square. The complete first-half certificate treats all 69 proper trace tags at this prime: their 51 distinct sums yield 507 eligible residuals, 478 nonnegative discriminants, and no integral target. This is a fixed-sum obstruction, not an ES counterexample and not an obstruction to the different factor-sum arrow.

A precise limitation on descent. The source-changing rules above do not apply to every trace. A separate, exhaustive standard-library certificate proves that 67369 is the least prime 1 mod 24 for which the first trace shell is earlier than every full shell. Its first trace has $a = 16849$, $R = 27$; its first full gate has $a = 16850$, $R = 31$. All 817 primes in the stated class through that bound and all their shells through first full occupancy are replayed by two implementations. Hence a universal repair from *every* trace cannot require $a' \leq a$. This is an obstruction to a proposed termination invariant, not an ES counterexample. Universal occupancy of the full original source is still unproved.

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